

• TRL 4-5 • BENCH VALIDATED

Modular Intramedullary Fixation System for Critical Long-Bone Defects

A temporary internal stabilisation system designed for staged reconstruction of critical bone defects with PMMA spacer compatibility.



Modular Architecture

Two-component system adaptable to defect size



PMMA Spacer Compatibility

Designed for antibiotic-loaded spacers



HA-Ag Antibacterial Coating

Selective coating in bone-contact zones



Adjustable Construct Length

Variable defect sizes accommodated

TREATMENT PATHWAY

External Fixation

Current standard



CITadel (TIS)

Temporary internal stabilisation



Definitive Reconstruction

Bone transport / graft

Current Limitations

- ✗ External fixation dependency
- ✗ No spacer-compatible internal systems
- ✗ Limited options for critical defects
- ✗ High complication rates

CITadel Advantages

- ✓ Internal temporary stabilisation
- ✓ Full PMMA spacer compatibility
- ✓ Defect-focused construct logic
- ✓ HA-Ag antibacterial interface

CLINICAL APPLICATIONS



Infected Bone Defects

Staged debridement with antibiotic spacer support and delayed reconstruction.



High-Energy Trauma

Blast injuries and segmental bone loss requiring temporary internal fixation.



Oncologic Resection

Post-resection defects needing stable fixation during staged limb salvage.



Revision Surgery

Complex cases where previous fixation has failed and staged approach is needed.

DEVELOPMENT STATUS

TRL 4-5

CURRENT LEVEL



PROTOTYPE TESTED



EARLY CLINICAL USE

→ TRL 6

TARGET: CLINICAL PILOT

CiTadel is designed as a temporary internal stabilisation system for critical long-bone defects during staged reconstruction procedures. The system complements existing intramedullary and spacer-based infection management solutions by providing stable modular fixation compatible with **PMMA spacers**. Patent pending.



Maksym Baida, PhD ORTHOPAEDIC TRAUMA SURGEON

Over 15 years of clinical practice in traumatology and reconstructive orthopaedics. Associate Professor at Bogomolets National Medical University, Kyiv.

Let's Talk

Open for collaboration with industry partners, clinical centres, and strategic stakeholders.



meredian18@gmail.com



+380 66 766 7094